Lesson 04: The Research Cycle and Finding a Research Topic

English Stage 4 (Psychology) · Unit 1: Psychology as a Science

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 2.1 Prelude to the Scientific Method; 2.2 A Model of Scientific Research in Psychology; 2.3 Finding a Research Topic.

In this lesson

You will follow the cycle of scientific research, learn where research ideas come from, and find out what counts as the research literature and how to search it. On the language side, you will study the tenses of an abstract and the passive voice for describing processes. You will then start your research proposal with a topic and a literature plan.

Learning goals

After this lesson you should be able to:

- Describe the cyclical model of research with the book's examples.
- Identify the three most common sources of research ideas.
- Distinguish sources that belong to the research literature from those that do not.
- Plan a literature search with databases, index terms and other techniques.
- Use present perfect, past and present tenses as an abstract does, and the passive to describe a process.

Key language

Term or phrase	Meaning	Example
research literature	all the published research in a particular field	Mehl's article became part of the research literature.
professional journals	periodicals that publish new research	*Psychological Science*; *Memory & Cognition*
empirical research report	a journal article in which the author reports on a new empirical study	Mueller and Oppenheimer's note-taking studies
review article	an article that summarizes previously published research, usually organizing or explaining it in a new way	A review that presents a new theory is a **theoretical article**; one that gives a statistical summary of all previous results is a **meta-analysis**.
peer review	new research submitted for publication is reviewed by two or more experts before an editor decides whether to publish it	In double-blind review, authors and reviewers do not know each other's identity.

Term or phrase	Meaning	Example
edited volume	a scholarly book with an editor or small group of editors and chapters by several different authors	Chapters may even disagree.
abstract	a short summary (about 200 words) of a research article	the laptop note-taking abstract in 2.1
literature review	a written summary of previous research on a topic	Reviewing the literature means finding, reading and summarizing it.
PsycINFO	the primary computer database that catalogs research in psychology	nearly four million records
index term	a standardized term that describes the content of each record in a database	"Human Sex Differences"; "Learning Strategies"
Probably the most common ... is ...	introduces the main item in a list	Probably the most common inspiration ... is previous research.
As a general rule, ...	gives advice that is usually right	As a general rule, use methods that others have used successfully.
by all means	certainly (strong encouragement)	"then, by all means, you should include it in your review"

Rules and explanations

The ideas

An abstract (2.1). In three studies, Mueller and Oppenheimer (2014) found that students who took notes on laptops did worse on conceptual questions than students who wrote by hand. Their abstract states a research question and why it is worth investigating: laptop note taking is ubiquitous and may harm learning.

The research cycle (2.2). The book's figure shows this cycle:

research literature → research question → empirical study → data analysis → conclusions → publication → research literature

Because published research is one of the main sources of new questions, research is a cycle. Questions can also start outside it, with informal observations or practical problems, but even then researchers first check the literature, to see if the question has been answered and to refine it.

- **Mehl's study:** the question came from stereotypes and published claims; the literature had not adequately addressed it; the team collected and analyzed data, drew conclusions and published, raising new questions (for example, about cultural differences).
- **Cell phones and driving:** earlier research showed that a simple verbal task impairs a perceptual or motor task done at the same time; new studies found that phone use impaired hazard detection,

reaction time and vehicle control, and each study joined the growing literature.

Where research ideas come from (2.3). Creativity is largely ordinary thinking strategies and persistence (Weisberg, 1993). Three common sources:

1. **Informal observations** of our own and others' behavior, or secondhand from newspapers, books and blogs. Milgram's research on obedience was inspired in part by reports of the trials of Nazi war criminals who claimed they were "only obeying orders".
2. **Practical problems**, leading to applied research: Does taking notes by hand improve exam performance? To what extent do cell phones impair driving?
3. **Previous research**, probably the most common source: ask an experienced researcher, or read the titles and abstracts in a journal.

The research literature consists almost entirely of articles in professional journals and scholarly books. It does not include self-help and pop psychology books, dictionaries, encyclopedias or websites for the general public, which are not reviewed by researchers and are often based on common sense or personal experience. Wikipedia is unsuitable: its authors are anonymous, may lack expertise, and its content keeps changing. Reviewing the literature early tells you whether your question has been answered, helps you judge how interesting it is, gives you ideas for your method, and shows how your study fits in.

Journals and books. Most journal articles are empirical research reports or review articles. In **double-blind peer review**, authors submit a manuscript to the editor, who sends it to two or three experts; they send back critical but constructive reviews, and the editor accepts the manuscript, asks for changes, or rejects it. Scholarly books (monographs and edited volumes) are usually peer reviewed too.

Searching the literature

- **PsycINFO** (from the APA) covers thousands of journals and books over more than 100 years. Each record gives publication information, an abstract, the works cited, keywords and standardized **index terms**, which its thesaurus helps you find.
- **Other techniques:** read the reference lists of good articles (a recent review is best); find works that cite a classic article; search the Internet and Google Scholar; ask people who know the topic.
- **What to look for:** sources that help you refine your question, choose methods, place your study in context and write your report. Start with the past five years (plus classic articles that nearly everyone cites); look for review articles, empirical reports whose methods you could use, and facts that show why your question matters.

Language focus

1. Tenses in an abstract (2.1)

Function	Tense	Example
the situation now	present simple	"Taking notes on laptops rather than in longhand **is** increasingly common."

Function	Tense	Example
what earlier research has done	present perfect	"Many researchers **have suggested** that laptop note taking is less effective"; "Prior studies **have** primarily **focused** on ..."
what this study did and found	past simple	"In three studies, we **found** that students who took notes on laptops **performed** worse ..."
the authors' claims	present simple, often hedged	"The present research **suggests** that ... they **may** still **be impairing** learning"

2. The passive for processes and systems. Use the passive when the process matters more than who does it:

- "The issues **are organized** into volumes."
- "Research on note-taking **is** always **indexed** under the term 'Learning Strategies.'"
- "the identities of the reviewers ... **are published** alongside the journal article"

Use the active when the actor matters: "The editor then **decides** whether to accept the article."

> **Note for Arabic speakers** - *Research*, *literature*, *evidence* and *information* are uncountable in English: "research on sleep", "a study", "two studies", not "a research" or "researches".

Practice

1. Reading questions

1. In the abstract in 2.1, what had prior studies focused on, and what new explanation does the present research offer?
2. Name the five steps of the model in 2.2. Why is it a cycle?
3. What inspired Milgram's research on obedience, and which source of ideas does this illustrate?
4. Why are pop psychology books and Wikipedia not part of the research literature?
5. Why is double-blind peer review helpful, and how does open peer review differ?
6. How does the PsycINFO thesaurus help in the "early memories" example?

2. Book exercises. Sections 2.1 to 2.3 have no exercises of their own; Chapter 2's exercises are in Section 2.9 (Lesson 06).

3. Language exercise

A. Name the tense of the verbs in bold and say why it is used.

1. "Prior studies **have** primarily **focused** on students' capacity for multitasking and distraction when using laptops."
2. "In three studies, we **found** that students who took notes on laptops **performed** worse on conceptual questions ..."
3. "We **show** that whereas taking more notes **can** be beneficial, laptop note takers' tendency to transcribe lectures verbatim ... **is** detrimental to learning."

B. Rewrite in the passive, leaving out the agent if it is not important.

1. The American Psychological Association produces PsycINFO.
2. The editor sends the manuscript to two or three experts.
3. The editor forwards the reviewers' comments to the researchers.

C. Write two sentences about your own topic in the style of an abstract: one on previous research (present perfect) and one hedged claim (a modal verb) about what your study will test.

Speak with your tutor

- "Here is my description of the Mehl study as a research cycle: [description]. Check that I named every step."
- "PSYC 110 Lecture 15 (Morality) says that the motivation for Milgram's work was the Holocaust. Ask me how this fits the book's idea of informal observations as a source of research ideas."
- "Act as a university librarian. Help me choose search terms and combinations for my topic, and explain the difference between keywords and index terms."
- "Name six sources (an encyclopedia entry, a meta-analysis, a self-help book, and so on). I will say which belong to the research literature and why."

Writing task

Research proposal, part 1: topic and literature plan. Write 250 to 350 words in two parts. The default topic is **sleep and memory in university students**, the topic you worked on in Stage 3; if you wrote an annotated bibliography then, start from it.

Part 1: Topic (one paragraph). Your research idea (a behavior or psychological characteristic and a group of people); where it came from (an informal observation, a practical problem or previous research); and why it matters. For example, PSYC 110 Lecture 17 mentions the idea that during sleep, memories "get played over and over again to consolidate them": an idea to look up in the research literature.

Part 2: Literature plan (a paragraph or a list).

- where you will search: PsycINFO, if a library gives you access; otherwise Google Scholar and open access journals such as *Frontiers in Psychology*;
- at least five search terms, from general to specific (for example, *sleep*, *memory*, *memory consolidation*, *naps*, *word pairs*), and two combinations (for example, *sleep* and *memory* and *college students*); note that you will check the thesaurus for index terms;
- what you will look for: a recent review article (past five years), two or three empirical reports whose methods you could use, a classic article that many sources cite, and facts that show why the topic matters (for example, how common short sleep is among students);
- which sources you will not use as evidence (pop psychology books, Wikipedia, websites for the general public), and how they may still help (their references).

Use the present perfect for previous research, *will* for your plans, and the passive where the process matters.

If you work on your own topic, write both parts for it in the same way.

Self-editing checklist:

- Does my topic name a behavior or characteristic and a group of people?
- Did I say which source of inspiration it came from?
- Did I explain why the topic matters?
- Did I list search tools, at least five search terms and two combinations?
- Did I plan to find a recent review, empirical reports and a classic article?
- Did I exclude pop psychology books, encyclopedias and Wikipedia as evidence?
- Did I use *research* as an uncountable noun and the present perfect for previous research?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "Understanding driver distraction", an APA video with David Strayer (Section 2.2): https://youtu.be/XToWVxS_9lA
- "How to Develop a Good Research Topic" (K-State Libraries, Section 2.3): https://youtu.be/nXNztCLYgxc
- "Using Google Scholar" (Section 2.3): https://youtu.be/t1ZwgDeX2eQ
- PSYC 110 Lecture 17 (sleep and dreams in Chapters 6 and 7, from 38:09): https://oyc.yale.edu/psychology/psyc-110/lecture-17
- Note: the book's sample PsycINFO search video is no longer available.

Answer key

1. Reading questions

1. On students' capacity for multitasking and distraction when using laptops. New explanation: even when laptops are used only for notes, they may impair learning because they lead to shallower processing: laptop note takers transcribe lectures verbatim instead of reframing the information in their own words (2.1).
2. Formulate a research question, conduct an empirical study, analyze the data, draw conclusions, and publish the results. It is a cycle because published research becomes part of the literature, which is one of the main sources of new questions (2.2, first paragraph).
3. Journalistic reports of the trials of accused Nazi war criminals, many of whom claimed that they were only obeying orders. It illustrates informal observations, here secondhand ones from newspapers (2.3, "Finding Inspiration").
4. They are not reviewed by other researchers and are often based on little more than common sense or personal experience. Wikipedia's authors are anonymous and may lack training or expertise, and its content continually changes (2.3).
5. It ensures that work meets the basic standards of the field before it enters the literature. Open peer review publishes the reviewers' identities alongside the article (they stay hidden during the review), for transparency and accountability (2.3).
6. "Memory" alone returns far too many records; the thesaurus suggests more specific index terms such as "early memories", and combining it with "human sex differences" returns fewer, highly relevant articles (2.3, "Using PsycINFO").

2. Book exercises. See Lesson 06.

3. Language exercise

A. 1. Present perfect: earlier research up to now. 2. Past simple: what this study did and found. 3. Present simple (with the modal *can*): the authors' general claims and conclusions.

B. 1. PsycINFO is produced by the American Psychological Association. (Close to the book's wording; the agent matters here.) 2. The manuscript is sent to two or three experts. 3. The reviewers' comments are forwarded to the researchers.

C. (example answer) "Previous studies have examined the role of sleep in memory. Sleep after learning may help students remember more, and the proposed study is designed to test this."

My notes